Real Time Rainfall Monitoring System

The project follows 13 weekly sprints from 28 July to 23 October 2026, with work scheduled Tuesday to Friday. Planning, development, bench testing, documentation and presentation preparation are Completed. Final deployment, baseline field calibration and the 48-hour field validation activities remain In Progress.

Product Backlog	In Progress	Completed
Sprint 1: Project Initiation & System Architecture		
Day 1: Define Project Scope & Board Setup		Completed
Day 2: Component Verification & Sourcing		Completed
Day 3: Development Environment Configuration		Completed
Day 4: Architecture Diagram & Data Flow		Completed
Sprint 2: Core Firmware & Sensor Interfacing		
Day 1: ESP32 Base Setup & Serial Output		Completed
Day 2: DS18B20 Temperature Sensor Logic		Completed
Day 3: TR-525M Rain Gauge Logic		Completed
Day 4: Firmware Formatting & Low Power Mode		Completed
Sprint 3: HiveMQ Cloud Broker Setup		
Day 1: Broker Provisioning		Completed
Day 2: Topic Structuring & Access Control		Completed
Day 3: Network Connection & Publish Testing		Completed
Day 4: Database Storage Integration		Completed

SAEON Product Backlog

Sprints 4 to 6

Product Backlog	In Progress	Completed
Sprint 4: Cellular Communication (SIM7000G)		
Day 1: SIM Module Wiring & Power		Completed
Day 2: Network Bearer Configuration		Completed
Day 3: MQTT over Cellular		Completed
Day 4: Stability & Error Handling		Completed
Sprint 5: SAEON Dashboard Base Application		
Day 1: App Skeleton & Routing		Completed
Day 2: Real-Time Visualization		Completed
Day 3: Historical Data & Filtering		Completed
Day 4: Cloud Command Prep		Completed
Sprint 6: Power Supply & Solar Integration		
Day 1: Battery & Charge Controller Setup		Completed
Day 2: Solar Panel Integration		Completed
Day 3: Mains Power Alternative Testing		Completed
Day 4: System Power Flow Audit		Completed

SAEON Product Backlog
Sprints 7 to 9

Product Backlog	In Progress	Completed
Sprint 7: Advanced UI & Health Diagnostics		
Day 1: Develop Station Health Diagnostics UI Component		Completed
Day 2: Implement CSV Export Functionality		Completed
Day 3: Mobile & Tablet Responsiveness		Completed
Day 4: Full End-to-End Data Pipeline Ping Test		Completed
Sprint 8: Hardware Finalization & PCB Soldering		
Day 1: Circuit Transfer & Layout		Completed
Day 2: PCB Soldering		Completed
Day 3: Enclosure Preparation		Completed
Day 4: Component Mounting & Sealing		Completed
Sprint 9: Pre-Deployment Auditing & Calibration		
Day 1: Power Drain Audit		Completed
Day 2: Solar Charging Verification		Completed
Day 3: Full System Deployment	In Progress	
Day 4: Baseline Calibration	In Progress	

SAEON Product Backlog

Sprints 10 to 12

Product Backlog	In Progress	Completed
Sprint 10: 48-Hour Field Stress Test		
Day 1: Stress Test Initiation	In Progress	
Day 2: Active Monitoring	In Progress	
Day 3: Manual Field Measurements	In Progress	
Day 4: Data Auditing & Conclusion	In Progress	
Sprint 11: Presentation Formatting & Content		
Day 1: Presentation Outlining		Completed
Day 2: Technical Diagrams & Architecture Slides		Completed
Day 3: Dashboard & Software Slides		Completed
Day 4: Technical Documentation		Completed
Sprint 12: Rehearsal & Submission Prep		
Day 1: Slide Deck Finalization		Completed
Day 2: Group Rehearsal 1		Completed
Day 3: Timed Rehearsal & Feedback		Completed
Day 4: Final Compilation		Completed

SAEON Product Backlog

Sprint 13

Product Backlog	In Progress	Completed
Sprint 13: Project Buffer & Final Review		
Day 1-4: Buffer & Prototype Documentation Sign-off		Completed

Status and source notes

The statuses below reflect the project owner's updates. Trello is the task-tracking board; GitHub stores code, documentation and evidence. The sprint backlog uses the same 13-sprint schedule and task statuses. These document updates do not modify the live Trello board.

Workflow: Product Backlog, In Progress and Completed. Sprint headings organise tasks and do not have completion marks. Sprint 13 sign-off covers the prototype documentation and presentation package; it is not final field-deployment acceptance. Supabase, historical data, CSV export, battery monitoring and solar integration are included in the current project scope.

Archived lists and archived cards are excluded. This product backlog shows sprint and daily card titles; individual checklist subtasks remain in Trello.

Source: SAEON Real-Time Rainfall Monitoring System Trello board export supplied by Ethan Williams.